

EX No:1a	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

Aim:

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications.

Algorithm:

Algorithm 1: Installing Python

1. Download Python from python.org
2. Run the installer
3. Select "Add Python to PATH"
4. Click Install Now
5. Verify installation using: `python --version`
6. Observe installed Python version

Algorithm 2: Installing Jupyter Notebook

1. Open Command Prompt
2. Install Jupyter `pip install notebook`
3. Launch Jupyter Notebook `jupyter notebook`
4. Browser automatically opens Notebook Dashboard.

Algorithm 3: Creating a Notebook

1. Open Jupyter Dashboard
2. Select New → Python 3
3. Create Notebook
4. Rename Notebook
5. Create Code and Markdown Cells
6. Execute cells using Shift + Enter

Program:

Program 1 — Simple Python Program

```
print("Welcome to Data Analytics Lab")
```

Output:

Welcome to Data Analytics Lab

Program 2 — Arithmetic Operations

```
a = 20
```

```
b = 10
```

```
print("Addition =", a + b)
```

```
print("Subtraction =", a - b)
```

```
print("Multiplication =", a * b)
```

```
print("Division =", a / b)
```

Output:

Addition = 30

Subtraction = 10

Multiplication = 200

Division = 2.0

Program 3 — Display Student Details

```
name = "John"
```

```
dept = "AI&DS"
```

```
cgpa = 8.9
```

```
print("Name:", name)
```

```
print("Department:", dept)
```

```
print("CGPA:", cgpa)
```

Output:

Name: John

Department: AI&DS

CGPA: 8.9

Program 4 — Creating Markdown Cell

```
# Data Analytics Lab
```

```
## Experiment 1
```

```
### Jupyter Notebook
```

Then press **Shift + Enter**.

Program 5 — Using NumPy Library

```
import numpy as np
```

```
a = np.array([10, 20, 30, 40, 50])
```

```
print(a)
```

```
print("Mean =", np.mean(a))
```

```
print("Sum =", np.sum(a))
```

Output:

```
[10 20 30 40 50]
```

```
Mean = 30.0
```

```
Sum = 150
```

Program 6 — Using Pandas Library

```
import pandas as pd
```

```
data = {
```

```
    'Name': ['John', 'David', 'Alex'],
```

```
    'Marks': [85, 90, 95]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

Output:

```
   Name  Marks
```

```
0  John    85
```

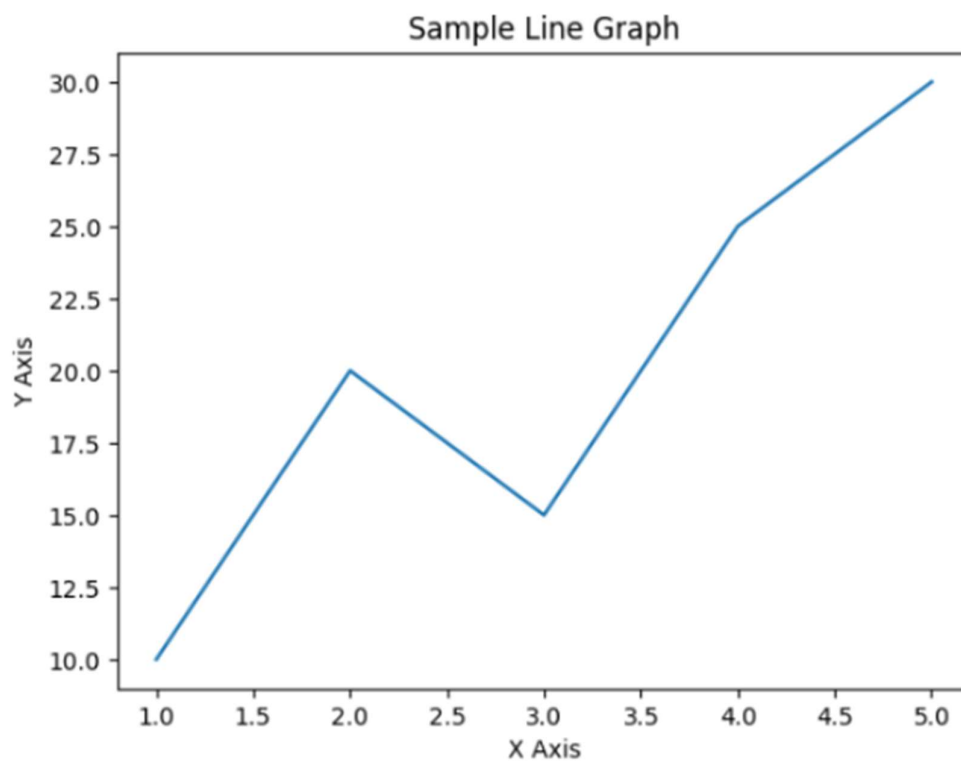
```
1  David    90
```

```
2  Alex    95
```

Program 7 — Data Visualization

```
import matplotlib.pyplot as plt  
x = [1, 2, 3, 4, 5]  
y = [10, 20, 15, 25, 30]  
plt.plot(x, y)  
plt.title("Sample Line Graph")  
plt.xlabel("X Axis")  
plt.ylabel("Y Axis")  
plt.show()
```

Output:



Result: